

The Triadic Measurement Substrate

$$\bar{B} = 3\bar{\alpha}$$

Volume 2

Paper 4: Time: The Now-Surface Deepened — Confirmed Past, Unwritten Future, and the Decay of the Wake

Aaron Switzer

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Paper 4

Time: The Now-Surface Deepened — Confirmed Past, Unwritten Future, and the Decay of the Wake

Aaron Switzer

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Status of Results

This paper continues the labeling conventions of Volume 1 and of the preceding papers of this volume. Every claim in the body falls into exactly one category, marked where it appears. No claim is upgraded beyond what its derivation earns.

Derived. Results that follow from the five axioms, the Principle of Generative Economy, and U_{sh} by explicit construction or proof, or from results already derived in Volume 1 and the earlier papers of this volume by explicit inheritance. The asymmetry between a confirmed past and an unwritten future as the two sides of the now-surface (§ I); the reconstructability of an interior confirmation's value from the conserved polarization it deposits into its forward cone, increasing with depth into the wake, with detection-only at the frontier (§ I); the fixed-fraction form of re-read decay, forced by the access constraint on the reinterpretation operator (§ IV, Appendix A); the monotone forgetting curve with a count-conserved floor (§ IV); and the structural flattening of the cosmic-residue balance (§ IV) are derived in this sense.

Derived-conditional. Results that follow given a primitive established elsewhere whose own grounding is stated. The per-fold shed value $s = 2$ is inherited from Paper 3, where it is forced given binary-agent triads; this paper uses the value and does not re-establish it.

Structural correspondence. Results in which a TMS construction reproduces the skeleton of a known framework or quantifies a recognized distinction, never used as support for a TMS claim, only as a landing point after one. The reading of re-read decay as lossy recompression in the rate-distortion sense (§ IV, § V); and the correspondence of the poised pre-confirmation configuration to an excitable system held near threshold (§ III, § V) are structural correspondences in this sense.

Predictions / Conjectures. Falsifiable claims that follow from the framework but whose empirical tests are not established here. That a memory degrades by the same law whether overwritten from the world or re-read from within — that memory decays through use, not only through neglect (§ III) — is a prediction in this sense; its match to data is not claimed.

Permanent open horizon. A question the framework is structurally unable to close, marked so that no later claim assumes it closed. What fills an unwritten interstice when it is finally confirmed — whether the content of the future has any determination prior to its folding — is not settled by anything in this paper and is left open (§ I, § II). This silence is load-bearing.

Deferred. Quantities the framework defines but does not evaluate here. The value of the cosmic residue-to-ordinary ratio depends on two rates this paper does not derive and is deferred (§ IV). The identification of the anchored residue with the dark-matter mode, and the mass–energy reading that binds it to dark energy, belong to the material side and are treated there, not here (§ IV).

Notation

Symbols are inherited from Volume 1 and the earlier papers of this volume. $\hat{R}$ is the reinterpretation operator (Volume 1, Definition 0.0.67); the now-surface is the active codimension-one confirmation frontier and the wake is the settled three-dimensional record behind it (Paper 1, § 1); U_{sh} is the maximum-entropy substrate field; s is the per-fold distinction shed of Paper 3, with $s = 2$ for binary-agent triads; n counts folds applied to a given structure; $r = 2^{-s}$ is the per-fold survival ratio introduced in § IV.

I. The Now-Surface Deepened

Paper 1 introduced the now-surface as the active frontier where agents are presently locking bits, and the wake as the settled record that accumulates behind it. That construction was developed entirely on the material side; it carried gravity, the black hole, and dynamic dark energy, and assumed no property of observers. This paper takes the same object and asks what it determines about time itself. The question is single: where, on the substrate, is the asymmetry between past and future, and what governs the relation of one now to the next.

The answer is already present in the three-state space of Volume 1. Paper 1 § 1.4 fixes three ontologically distinct states for an interstice: latent, drawn from U_{sh} and neither zero nor one but the incompressible superposition of both; confirmed-zero; and confirmed-one. Confirmation resolves the latent state into one of the two confirmed values, and by Axiom 4 the result cannot be overwritten. These three states, read against the now-surface, are the three regions of time.

The confirmed wake is the past. It is the locked record, sealed by Axiom 4: the system has only one available direction, and each flop adds a layer without disturbing prior layers. The now-surface is the present: the codimension-one frontier where confirmation is currently occurring, the only place a fold can fire. The future is the latent population the present surface has not yet folded. The identification is relative to a surface and not global: latency is a property an interstice has at large, but it reads as future only with respect to a given now-surface, for which it is the not-yet-confirmed region that surface can advance into. What is future to one frontier is unrelated to another that will never reach it. So read, the future is unwritten in the strict sense: not merely unknown but undetermined, the superposition not yet resolved into a confirmed value.

This locates the asymmetry exactly. The past and the future are not two symmetric directions along a line; they are two different states of an interstice relative to a single surface. The past is what has been confirmed and sealed; the future is what remains latent; the now-surface is the boundary between them, and it advances under $\hat{R}$ every flop (Paper 1, § 7). The arrow of time is not added to this picture. It is Axiom 4 read as a statement about which side of the now-surface a fold can act on. A fold acts in exactly two ways. It can confirm a latent interstice, resolving the superposition into a confirmed value and moving the interstice from the future side of the surface to the past side. It can also re-read a region of wake already confirmed. What it cannot do is return a confirmed interstice to the latent state: that is precisely the overwrite Axiom 4 forbids. Confirmation therefore runs one way across the surface, and the one-way-ness is not a separate postulate but the causal anchoring of Volume 1 read at the surface.

Two consequences follow immediately, and they organize the rest of the paper. First, only the past decays. A latent interstice carries no confirmed distinction to lose; there is nothing in the future to shed, because

nothing there has been written. Degradation is therefore a strictly past-facing phenomenon: it acts on the wake, never on the latent population. The temporal arrow and the direction of decay are the same fact — one can re-read the wake and lose grain, but one cannot re-read the unwritten, because there is nothing yet to read. Decay is developed in § IV.

The asymmetry is directional, not an asymmetry of determination. The substrate geometry runs both ways across the surface, and the difference between past and future is the difference between running it backward into the wake and running it forward into the latent region. Run backward, the geometry reconstructs, and it does so for a reason internal to the substrate. Each confirmation transports its polarization outward through the $1 \rightarrow 4$ multiplier: a confirmed value emits its signals into the downstream synthetic bits, which carry that value's polarization forward, and by Axiom 3 those signals are conserved and cannot be lost (Paper 1, § 4.2–4.3). A confirmation in the interior of the wake has therefore already deposited its value, redundantly, across the layers it produced. If a single agent's contribution is later disrupted, the local unit only registers that a confirmation was present — the boundary opens and a parity syndrome appears — but the original value remains recoverable from the conserved polarization the confirmation laid into its forward cone. The past is rigid because it is over-determined by its own downstream record, and the over-determination grows with depth: the deeper a confirmation lies in the wake, the more forward layers redundantly encode it, so older structure is more strongly fixed than recent structure. This redundancy structure corresponds to the $[[4, 2, 2]]$ error-detection code, whose single-error detection sits on the local unit and whose reconstruction is carried by the forward cone (Paper 1, § 5); the correspondence is noted as a landing, not relied on as the argument.

The reconstruction has one limit, and it falls exactly at the present. A confirmation at the now-surface frontier has not yet produced a downstream cone, so a disruption there can only be detected and re-confirmed against the current polarization landscape, which biases the redrawn value without fixing it. Rigidity of value is a property of the wake interior; at the frontier only the fact of confirmation is preserved, not its value. This is consistent with the asymmetry as a whole: the past is rigid in proportion to how much wake lies downstream of it, and the present, having none, is the one place the record is not yet fixed.

Run forward, the same geometry is blocked, and by two obstructions of different kinds. The first is the latent draw itself: the next confirmation resolves against a fresh contribution from U_{sh} , which is maximum-entropy and has no attractors (Paper 1, Axiom 0; § 7.2), so the forward step is not determined by present structure however completely that structure is known. This obstruction is ontological and holds for any observer. The second is positional: a bounded subsystem cannot measure across its own closure boundary, and the exterior it cannot read co-determines the folds it will undergo, so from inside the subsystem the forward step is unreadable even where it is determined. This obstruction is relative to the observer's position with respect to the boundary; an observer outside the system would not encounter it. The forward direction is therefore unshortcuttable from within, which is the time-like face of the record's computational irreducibility (Paper 1, § 7.2a). The past is reconstructable and the future is not, from one geometry, because reconstruction runs with the closed constraints and the conserved downstream record, while prediction runs against an undetermined draw and an unreadable exterior.

Second, the future is not modeled by reaching toward it. Nothing on the present surface extends into the latent region or carries information about which value an interstice will take. What a system can do is fold on its own wake — on structure already written — to build a present configuration, and that configuration can bias which latent interstices the surface confirms next without containing any of their content. The configuration is past-built and present-held; it points at the future only in that the world may or may not later confirm it. This is the subject of § II. What fills a latent interstice when it is finally confirmed is not

determined by anything on the present surface, and this paper does not claim it is; that silence is marked as a permanent open horizon and is not crossed.

The paper proceeds as follows. § II develops the present configuration poised over an unwritten interstice and the construction of models from the wake. § III derives the re-read clock that carries distinction across nows and states the prediction that re-reading degrades. § IV derives the decay law itself, in both its forgetting and cosmic-residue readings, and fixes what it leaves open. § V situates the paper in the literature. The Conclusion forwards nothing beyond what is derived.

II. The Excitable Manifold and the Construction of Models

§ I left the future strictly unwritten and ruled out any reach from the present surface into the latent region. This raises a structural question. A confirmation event requires triadic agreement among agents before a latent interstice resolves (Paper 1, § 1.4); between events, the agents bounding an unconfirmed interstice are not folding, yet they are not inert either, since the next fold proceeds from their current configuration. The question is what to call that state without importing a relation to the future the substrate does not have.

The agents themselves supply the constraint. By Paper 1 § 1.2 the agent manifold is immutable: agents do not change, move, or degrade. Whatever the pre-confirmation state is, it cannot be a state of an agent, because agents have no states to occupy. It is instead a property of the configuration: three agents and their touch-vector signals standing in a relation from which a fold can proceed, but in which triadic agreement has not yet been reached. The configuration is structurally complete and causally quiet. It does nothing until the conditions for agreement are met, at which point it folds.

This is the structure of an excitable system: a configuration that holds at a threshold and does not act until input crosses it, after which it fires (van der Pol & van der Mark, 1928; FitzHugh, 1961). We adopt the term for the pre-confirmation configuration: the excitable manifold is the region of agents poised to confirm a latent interstice that has not yet folded. The term is chosen for its restriction. An excitable configuration carries no model of what will cross its threshold and no relation to the value the fold will produce; it carries only a release condition. Naming the state this way records what is present — a threshold and the readiness to fire — and withholds what is not: any orientation toward the future, any anticipation, any holding-toward. The configuration is poised, not expectant.

What looks, from outside, like a system directed at the future is a present operation on the past. A subsystem can apply folds to its own wake — to structure already confirmed and sealed — and assemble from it a present configuration: a confirmed arrangement whose touch-vectors point forward into the lattice (Paper 1, § 1.3, forward propagation only). Such a configuration is a model in the only sense the substrate allows: a present, fully confirmed object, built from past material, that biases which latent interstices the now-surface is positioned to confirm next. It selects where the manifold is excitable. It does not contain the content of any future confirmation; it cannot, because that content is latent and undetermined until folded. The model is past-built and present-held, and it points at the future only in that the world may or may not later confirm the interstices it has made the system ready for.

Two boundaries close the section. First, the construction of models is a re-use of the wake, and re-use is a fold; by § IV it sheds. A system that builds richly from its past degrades that past in the building, by the same law that governs all re-reading. The model-building capacity and the decay of the material it builds from are not separate facts. Second, what fills a latent interstice when the excitable manifold finally folds it is not supplied by the model, by the configuration, or by anything on the present surface. The model fixes *where* the system is ready to confirm; it does not fix *what* is confirmed there. Whether the content of the future has any

determination prior to its folding is left open, exactly as § I marked it, and nothing in this section assumes an answer.

III. The Re-read Clock

The previous sections concern a single now and the latent region ahead of it. Time requires the relation between nows, and that relation is carried by re-reading. A re-read is one application of $\hat{R}$ to a region of wake already confirmed — the now-surface, advancing, folds a second time on structure it or an earlier surface has written. Let n count the folds a given region has undergone since it was written, and let f be the rate at which those re-reads occur, the re-reads per unit flop-time for that region. We call f the re-read clock. It is the variable that turns the single-now quantities of Paper 3 into across-now quantities: the form-surplus that Paper 3 sizes at one now is shed once per fold, so n folds carry it to r^n of its value, with r the per-fold survival ratio derived in § IV. The clock counts; § IV gives what each count costs.

A fold can be triggered from either side of the master equation, and the clock has a corresponding pair of faces. A re-read induced from the bit side — a stressor, sensory load, the world writing over a region — advances the clock by overwriting; call its rate f_{world} . A re-read induced from the agent side — the system folding on its own wake, the model-construction of § II — advances the same clock by re-confirmation; call its rate f_{self} . The two are faces of one operation: $\hat{R}$ does not record which side triggered it, and the distinction shed per fold is the same in both cases (§ IV). The total re-read rate is $f = f_{\text{world}} + f_{\text{self}}$. The agent-side face is the same operation named, in plain register, as recollection: to re-read a region of one's own wake is to fold it again.

One consequence follows directly and is falsifiable. Because the shed is the same whichever face advances the clock, a region re-read from the agent side degrades by the same law as a region overwritten from the world side. Re-confirmation is not a free replay of a stored value; it is a fold, and a fold sheds (§ IV). A memory therefore decays through use, not only through neglect: each recollection re-reads the trace and sheds from it, so a frequently re-read region loses fine distinction faster than one left alone, and what is recalled is increasingly the most recent re-reading rather than the original confirmation. This is a prediction; it is stated here as a consequence of the shared shed and is not matched to data in this paper.

The clock is a rate, and the present paper needs nothing more of it. Which region a re-read falls on is set in part already and in part downstream. When the trigger is from the bit side, the world sets the region: a stressor or sensory load re-reads whatever it falls on. When the trigger is from the agent side, the model of § II biases the region, since a configuration built from the wake makes some interstices excitable rather than others. What remains — what determines that f_{self} folds toward structure that persists across folds rather than structure that does not — is derived in the next paper, and it is derived there on the shed this paper establishes: unbound structure degrades under the per-fold shed of § IV into a configuration that cannot re-confirm, and that degradation is the machinery the next paper's selection rule runs on. The selection is therefore not left open but located: it is the agent-side dynamics of Paper 5, for which § IV supplies the decay. Here f enters only as a rate.

IV. The Decay Law

This section fixes what one fold costs and what many folds accumulate. The per-fold shed is inherited: Paper 3 establishes that the confirmation fold sheds s bits of distinction, with $s = 2$ for binary-agent triads, forced there and used here without re-derivation (Paper 3, § IV, Appendix B). What this section adds is the

form decay takes across folds, and it turns on a single question the substrate settles: whether the shed removes a fixed quantity of distinction per fold or a fixed fraction of it.

The question is decided by what $\hat{R}$ can act on. A re-read operates on the region as it currently stands, not on the original confirmation: by Axiom 4 the original is sealed in the wake, and the re-reading surface reads the present, already-degraded copy. A fixed-budget shed — removing s bits of the original total each fold — would require the fold to act against that original total, a quantity the surface no longer has access to. A fixed-fraction shed — removing a set proportion of the distinction currently present — requires only the present region, which is what the surface has. The fixed-fraction reading is therefore forced, not chosen; the access constraint admits no other. The per-fold survival ratio follows as $r = 2^{-s}$, and with $s = 2$, $r = 1/4$. The derivation, with the fixed-budget reading shown to fail, is given in Appendix C.

Accessible distinction after n folds then decays geometrically from its initial value toward a floor:

$$D(n) = D_{\text{floor}} + (D_0 - D_{\text{floor}}) r^n.$$

The floor is not zero. By Axiom 3 the fold conserves count while shedding distinction, so what survives unconditionally is the count-level record that the region was confirmed — the gist, in the agent-side register — while the fine distinction above the floor decays by r^n . The curve is monotone decreasing in n , its form is fixed by the access constraint rather than by a fitted rate, and it bounds below at $D_{\text{floor}} > 0$: detail is lost toward gist, never to total erasure, which would require the count loss Axiom 3 forbids. The three conditions are verified in Appendix C. The fixed-budget reading fails the third: a fixed subtraction drives D through the floor to zero in finite folds, erasing the count. That the substrate rejects fixed-budget for fixed-fraction is the same fact as the existence of the floor.

The same per-fold shed governs a second process, at the scale of the wake's own accumulation. A forming subsystem boundary anchors residue from the confirmed past at a rate A , and re-reads de-anchor it: by the law just fixed, each re-read sheds the fraction $(1 - r)$ of the residue currently anchored, at the clock rate f . The anchored residue R then obeys

$$\frac{dR}{dt} = A - f(1 - r)R,$$

a balance with a stable fixed point at $R^* = A/[f(1 - r)]$. The loss term carries the same $s = 2$ as the forgetting curve, in the same fixed-fraction sense, and reduces the residue rather than growing it. Anchored residue therefore self-limits rather than accumulating without bound, and the residue-to-ordinary ratio flattens as a structural consequence rather than by tuning. The sign, the shared s , and the commensurability of the two rates are checked in Appendix C.

The two processes are one law at two addresses, and the identity is sharp: the r of the forgetting curve and the r of the residue balance are the same $1/4$, used the same way — fixed-fraction — for the same reason, the access constraint that forbids a fold from reaching the original behind the present copy. Forgetting re-reads an anchored memory; the residue balance re-reads the accumulated wake; both are $\hat{R}$ advancing on what is already written, shedding the same fraction per fold. This is the unification, and it holds structurally rather than by coincidence of magnitude. The check that the two uses match in number and in kind is the decisive one and is carried in Appendix C.

What this section does not fix is also stated. The value of the residue-to-ordinary ratio is not derived: R^* depends on A and f , neither of which this paper derives, so the flatness is structural but the magnitude is open. The identity of the anchored residue with the material dark-matter mode, and the mass–energy reading that would bind it to the dark-energy residual, belong to the material side and are treated there (Volume 1, Paper 5); they are not claimed here. And whether either curve matches measured data — the forgetting

curve against recollection, the flattening against the observed ratio history — requires outside data and is not asserted. The framework fixes the form; the world is left to confirm it.

V. Where It Sits

The closest metaphysical neighbor is the growing block view of time, on which the past and present exist while the future does not, and the present is the surface at which new moments are added to the past, so that the block of what exists grows (Broad, 1923; Tooley, 1997; Forrest, 2004). The structure derived here is that view's structure: the wake is the existing past, the now-surface is the surface of addition, and the latent region is the future that does not yet exist. The fork is in standing, not in shape. The growing block posits the asymmetry as a metaphysical primitive; here it is derived — the now-surface is a forced structural object of the substrate (Paper 1), and past, present, and future are read off the confirmed, active, and latent states rather than asserted. The variant on which the past is sealed and only the present is active (Forrest's dead-past version) corresponds to the now-surface being the sole site of confirmation while the wake is locked record, with the qualification that this paper makes no claim about where measurement is occupied (§ I; the band-word discipline of Paper 3). The account departs from eternalism, on which all times are equally real, since the future is latent and undetermined rather than real elsewhere; and from presentism, on which only the present exists, since the wake is real and sealed.

The closest live interlocutor on the agent side is predictive processing, on which perception is the brain's top-down prediction of its input and learning is the minimization of prediction error against a generative model (Rao & Ballard, 1999; Friston, 2010). The fork is precise and is not a disagreement about data. On that account the present state carries a model *of* the future, projected forward; on this one the present surface carries no content about the future at all, because the future is latent and undetermined until folded (§ I), and what looks like prediction is a present configuration built from the wake that biases where the manifold is excitable (§ II). Prediction places the future inside the present as a model; this account keeps the future unwritten and the present merely poised. Any predictive-processing-like content, where it exists, would live in the model-construction of § II, not in the poised state itself. The exclusion-style divergence with integrated information theory is the subject of Paper 2 and is not reopened here.

The decay law has its nearest quantitative neighbors in two fields. In continual learning, sequential training induces catastrophic forgetting, and the retained-memory signal decays exponentially under ordinary gradient descent, taking a consolidated, slower form when the weights encoding old tasks are protected (Kirkpatrick et al., 2017); models that combine decay with stochastic reactivation reproduce smoothly decaying retention curves rather than a sharp cutoff. The exponential-toward-a-floor form of § IV is the same shape, and the reactivation that degrades while it consolidates is the agent-side re-read of § III seen in a network; the correspondence is a convergence, landed after the derivation, and the fork is that the law here is forced by one access constraint rather than fitted as a mechanism per task. Underneath both sits the thermodynamic floor: a logically irreversible, many-to-one operation reduces degrees of freedom and dissipates, at least $kT \ln 2$ per erased bit (Landauer, 1961; Bennett, 1982). The confirmation fold is exactly such a map — three loci to one bit — so the shed is the substrate's Landauer cost, and the decay law is that cost paid once per re-read. This is noted as the physical grounding of the shed, not as an independent result.

The poised pre-confirmation configuration corresponds to an excitable system held below threshold (van der Pol & van der Mark, 1928; FitzHugh, 1961), and the broader observation that information-bearing systems are often tuned near a critical or metastable point (Bak, Tang, & Wiesenfeld, 1987; Beggs & Plenz, 2003) is a convergence on the present construction, landed after it and not used to support it: a system poised at threshold is where its dynamic range is widest, which is the same content the now-surface carries on the

material side. The decay law also corresponds to lossy recompression in the rate-distortion sense (Shannon, 1948): re-reading is re-encoding through a channel that cannot hold the source's full distinction, and the gist floor is the rate below which the code cannot compress without losing the count. The generational loss of a copy made from a copy is the same structure observed outside the substrate; it is noted as a correspondence, not a citation of support. The reconstructability of the past from its forward record corresponds to the recent result that an error-corrected history's causal memory either propagates forward or terminates at detection (a structure independently described for monitored many-body systems); the depth-dependent form given here, in which rigidity grows with the wake downstream of a confirmation, is original to this paper. On the open-future question, the account sides structurally with the view that propositions about the unwritten future lack a determined value until the event (the future-contingents tradition, Aristotle, *De Interpretatione* 9), but reaches it from the substrate rather than from semantics: the value is undetermined because the latent state is, not because of a convention about truth. The observer-relative appearance of unpredictability — determined from outside a boundary, unreadable from within it — is the substrate's account of why sensitive dependence presents as it does to an embedded observer, and is offered as that, not as a contribution to dynamical-systems theory.

Conclusion

Time, on the substrate, is the now-surface read on both sides. The past is the confirmed wake, sealed by causal anchoring and reconstructable from the record it has already written, the more rigidly the deeper it lies. The future is the latent region the surface has not folded, unwritten in the strict sense and reachable only as a present configuration built from the past. The present is the one surface between them, where confirmation fires and where, alone, the record is not yet fixed. The asymmetry of past and future is not an asymmetry of determination but of direction across that surface: the same geometry reconstructs backward and is blocked forward, by an undetermined draw and an unreadable exterior.

What relates one now to the next is re-reading, and re-reading sheds. The per-fold shed of Paper 3, carried across folds by the re-read clock, fixes a single decay law that governs the fading of a memory and the self-limiting of the wake's residue alike, by one fraction used one way. The law fixes the form of both and the value of neither; what is derived is held apart from what is left open, and the open quantities are named so that they cannot later be assumed closed.

The re-read clock has been developed here only as a rate. Its agent-side face folds on the wake and sheds by the same law as the world's overwriting. What determines that this face folds toward structure able to persist across folds is taken up in the next paper, where the agent-side dynamics are derived — and derived on the shed established here, since it is the per-fold degradation of § IV that turns unbound structure into a configuration that cannot re-confirm. This paper supplies the decay; the selection that runs on it is the subject of the paper that follows. Here the clock turns, and what it turns on decays.

Appendix A: Reconstructability of the Wake and Its Depth-Dependence

This appendix establishes the claim of § I that a confirmed value is reconstructable from the record its confirmation deposits downstream, that the reconstructability increases with depth into the wake, and that it fails at the now-surface frontier. The objects are inherited from Volume 1; no new primitive is introduced.

A.1 The objects

A confirmation event at a region resolves a latent interstice to a value $\bar{B} \in \{0, 1\}$ with polarization $\chi \equiv 2\bar{B} - 1 \in \{-1, +1\}$. By the $1 \rightarrow 4$ multiplier (Volume 1, Paper 1, § 4.2), the event emits twelve outgoing signals consumed by four downstream synthetic bits, each carrying the source polarization χ (Paper 1, § 4.3). By Axiom 3 these signals are conserved: not created, destroyed, or deferred. Let $d \geq 0$ index depth into the wake, the number of confirmed layers built downstream of the event since it was written; $d = 0$ is the now-surface frontier.

A.2 The local unit detects but does not reconstruct

Disrupt one agent's contribution to the confirmed interstice. The agent is immutable (Paper 1, § 1.2); what is disturbed is the confirmation relation, opening the closed boundary and producing a parity syndrome (Paper 1, § 5.2). The syndrome establishes that a confirmation was present and that it was disturbed. It does not, on the local unit alone, fix the disturbed value: a 2-simplex with one vertex contribution missing is consistent with either χ , since detection is of the violation, not of the original polarization. The local unit yields detection only.

A.3 The forward cone reconstructs, for $d \geq 1$

For $d \geq 1$ the event has emitted its polarization into the downstream layers, where by A.1 the signals carrying χ are conserved. The original χ is therefore recoverable as the polarization carried by the conserved downstream signals the event itself produced, independently of the disrupted local contribution. The number of downstream signals carrying χ grows with the forward cone: each layer re-emits under the same multiplier, so the count of conserved carriers of the original polarization is non-decreasing in d . Reconstruction is thus available for $d \geq 1$ and is increasingly over-determined as d grows — the deeper the event lies in the wake, the more independent downstream carriers fix its value. This is the depth-dependence asserted in § I.

A.4 The frontier limit

At $d = 0$ no downstream layer has yet formed, so no conserved carrier of χ exists outside the disrupted unit. By A.2 the local unit yields detection only; re-confirmation must redraw the value against the current polarization landscape, which biases the redraw (Paper 1, § 4.3; Volume 1, Paper 2) without fixing it. At the frontier, therefore, only the fact of confirmation is preserved, not its value. Reconstructability of value is a property of the wake interior.

A.5 Break-test

The result fails if reconstruction is claimed where no forward cone carries the polarization. Two pre-registered failure conditions: (i) if the $1 \rightarrow 4$ multiplier were content-neutral — if downstream signals did not inherit χ — then no downstream carrier would fix the value and the claim would reduce to detection at all depths; Paper 1 § 4.3 establishes the inheritance, so this does not obtain. (ii) If the claim were extended to $d = 0$, it would assert reconstruction with zero carriers, which A.4 forbids; the claim is therefore restricted to $d \geq 1$ and is false outside that range, as required. The correspondence to the $[[4, 2, 2]]$ code (distance-2 detection on the local unit; reconstruction via the forward cone) is consistent with this structure but is not used in the proof.

Appendix B: The Forward Step Is Unreadable from Within the Boundary

This appendix establishes the second forward obstruction of § I: that a bounded subsystem cannot read the contribution its exterior makes to the folds it will undergo, so that even a determined forward step is unreadable from inside. It is distinct from the first obstruction, which is the genuine undetermination of the latent draw (Axiom 0) and holds for any observer.

B.1 The objects

A subsystem is a closure: a region individuated by a boundary across which it is defined as interior against exterior (Volume 1, Paper 3; Paper 5). A confirmation on the boundary resolves against contributions from both sides, since the now-surface advances on the whole local configuration, interior and exterior alike (Paper 1, § 1).

B.2 The argument

A measurement available to the subsystem is a confirmation it participates in — one whose loci lie within its closure. The exterior, by definition of the boundary, contains loci the subsystem does not enclose and in whose confirmations it does not participate. Their values are therefore not among the quantities the subsystem can read. A forward fold on the boundary resolves partly against those exterior contributions. Hence the subsystem cannot, from within, fix the exterior input to its own next fold, and the forward step is unreadable from inside to exactly the extent that it depends on the exterior — even when that step is fully determined from a vantage that encloses both sides. The obstruction is positional: it is a consequence of which loci fall inside the boundary, not of any indeterminacy in the dynamics.

B.3 Relation to the first obstruction and to irreducibility

The two obstructions are independent. The first removes determination at the source (the latent draw is undetermined, Axiom 0); the second removes readability from a position (the exterior is determined but unenclosed). Either alone blocks forward reading; together they fix that prediction from within runs against both an undetermined draw and an unreadable exterior. The positional obstruction is the time-like reading of the record's computational irreducibility (Volume 1, Paper 1, § 7.2a): the record cannot be compressed, and the forward step cannot be shortcut from within.

B.4 Break-test

The claim fails if an interior measurement could fix an exterior locus. By B.1 the boundary is defined precisely by non-enclosure of the exterior, so an interior confirmation enclosing an exterior locus would contradict the definition of the closure; no such measurement exists for a genuine boundary. The claim is correspondingly vacuous for a system with no exterior, which is the total-system case treated separately (Paper on the total system); there the obstruction does not arise because there is no unenclosed remainder, consistent with the result being positional.

Appendix C: The Decay Law

This appendix derives the per-fold survival ratio, the forgetting curve and its three conditions, the cosmic-residue balance and its sign and fixed point, and the unification check that the two uses share one ratio used one way. The per-fold shed s is inherited (Paper 3, Appendix B), with $s = 2$ for binary-agent triads.

C.1 Fixed fraction is forced; the survival ratio

A re-read applies $\hat{R}$ to a region as it currently stands. By Axiom 4 the original confirmation is sealed and cannot be re-accessed; the re-reading surface acts on the present copy. A fixed-budget shed removes s bits of the original total per fold and so requires the original total as an operand, which the surface does not possess. A fixed-fraction shed removes a set proportion of the distinction presently accessible and requires only the present region. Only the latter is a substrate-available operation; the fixed-fraction reading is forced. The per-fold survival ratio is $r = 2^{-s}$, and for $s = 2$, $r = 1/4$. The value tracks s : it is fixed by the inherited shed, not chosen to resemble decay.

C.2 The forgetting curve and its three conditions

With a fixed fraction r surviving per fold, accessible distinction after n folds is $D(n) = D_{\text{floor}} + (D_0 - D_{\text{floor}})r^n$. (A1) Monotone: $r \in (0, 1)$ gives r^n strictly decreasing, so $D(n)$ decreases monotonically toward D_{floor} ; shed distinction is not regained, since Axiom 4 seals the original. (A2) Derived form: the exponential form follows from the fixed-fraction result of C.1, and the rate $r = 2^{-s}$ from the inherited s ; no rate is fitted. (A3) Floor: by Axiom 3 the fold conserves count, so the count-level record (gist) is never shed; $D_{\text{floor}} > 0$ and the curve distinguishes detail-loss from total erasure. The fixed-budget reading fails A3: a fixed subtraction reaches and crosses zero in finite folds, destroying the count, which Axiom 3 forbids. The floor existing and fixed-fraction being forced are the same fact.

C.3 The cosmic-residue balance

A forming boundary anchors residue at rate A and de-anchors it by re-reading at clock rate f , each re-read shedding the fraction $(1 - r)$ of the residue then anchored. Hence $dR/dt = A - f(1 - r)R$. (B1) Sign: for $R, f, (1 - r) > 0$ the loss term is negative, reducing the residue, opposing accumulation. (B2) Same shed: $(1 - r) = 1 - 2^{-s}$ with the same $s = 2$, giving $3/4$. (B3) Units: A is residue per time; $f(1 - r)R$ is (folds per time)(dimensionless)(residue) = residue per time; commensurable, so cancellation is well-posed. The fixed point $R^* = A/[f(1 - r)]$ is stable, since $\partial_R[A - f(1 - r)R] = -f(1 - r) < 0$; the residue self-limits and the ratio flattens structurally.

C.4 The unification check

The decisive condition (X1): the r of C.2 and the r of C.3 must be the same value used the same way. Both are $r = 2^{-s}$ with $s = 2$; both enter as fixed fractions, not fixed budgets; and both are fixed-fraction for the same reason — the access constraint of C.1, which forbids a fold from reaching the original behind the present copy, in the memory region and in the accumulated wake alike. The unification is structural, not a numerical coincidence between two independently tuned processes.

C.5 What is not fixed; break-test

The value of R^* depends on A and f , neither derived here; the flatness is structural and the magnitude is open. The break-test is X1 itself: had forgetting forced fixed-fraction while the residue balance forced fixed-budget, or had the two required different s , the single-law claim would be false even with each half internally sound. Both forced fixed-fraction from one access constraint and share $s = 2$; the test is passed. A second pre-registered failure: if accessible distinction were defined against the sealed original rather than the present copy, C.1 would admit fixed-budget and A3 would fail; Axiom 4 forbids that access, closing the route.

Acknowledgments

This paper develops the now-surface of Paper 1 and the per-fold shed of Paper 3, and inherits its primitives from Volume 1. It is one paper of a larger derivation campaign and does not stand alone.

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